

PVT

掌锐电子

刷机后无网络问题分析

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问题概述

- 问题背景：
 - 线刷升级后，状态栏tbox一直无网络状态
- 问题现象：
 - 线刷升级后（C5L0CN330SU1F082903DEV升级到C5L0CN330SD1F083001DEV），观察状态栏tbox
 - 状态栏tbox一直为无网络状态

问题分析

- 分析日志：TBOX\log\mtklog\mobilelog\APLog_2020_0101_000003_1

从Tbox端日志看，CUXV5-33265中第一次开机log：APLog_2020_0101_000006__1，有发现两个boot__normal目录，可以推断出tbox在启动过程中，又发生了一次重启。

GwmLog_2025_08_30_09_16_17 > TBOX > log > mtklog > mobilelog > APLog_2020_0101_000006__1 >				
名称	修改日期	类型	大小	
boot__normal	2025/8/30 15:48	文件夹		
Last_1_boot__normal	2025/8/30 14:25	文件夹		
boot_time.svg	2025/8/30 1:25	Microsoft Edge ...	554 KB	
bootprof	2025/8/30 1:25	文件	3 KB	
bsp_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	18 KB	
ccci_dpmaif_debug_2020_0101_0000...	2025/8/30 1:25	CURF 文件	1 KB	
crash_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	1 KB	
events_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	1 KB	
kernel_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	286 KB	
main_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	5,174 KB	
mblog_history	2025/8/30 1:25	文件	5 KB	
pl_lk	2025/8/30 1:25	文件	24 KB	
properties	2025/8/30 1:25	文件	18 KB	
radio_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	1 KB	
reboot-reason	2025/8/30 1:25	文件	2 KB	
security_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	1 KB	
stats_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	1 KB	
sys_log_2020_0101_000009.curf	2025/8/30 1:25	CURF 文件	1 KB	

上一次开机的过程中，开机log还没有copy到SD卡or EMMC中，又发生了重启，上一次开机log就保存在这个文件夹，Last_1_boot__normal是上一次开机log。

问题分析

- 分析日志: TBOX\log\mtklog\mobilelog\APLog_2020_0101_000006_1\Last_1_boot_normal

从Last_1_boot_normal第一次开机log看, apn db数据库初始化过程还没完成就发生重启, apn adb数据库生成过程重启。

```
\MT8676\nobo\CUXV5-33265\GwmLog_2025_08_30_09_16_17\TBOX\log\mtklog\mobilelog\APLog_2020_0101_000006_1\Last_1_boot_normal\main_log_2010_0101_000046.curf
2020-01-01 00:00:00.033791    2486 D mtktelephonyser: 0x7fbeb4-0x ( 2486, 2498), [RADIO], [3], [Tag]RPApnController[0][TAG]: onInit
2020-01-01 00:00:00.036106    2486 D mtktelephonyser: 0x7fbeb4-0x ( 2486, 2498), [RADIO], [3], [Tag]RPApnController[1][TAG]: onInit
```

从boot_normal第二次开机log看, 加载到apn db数据库后, 读取adb db的数据发现文件已损坏, 数据异常, mtktelephonyservice读取apn list时发现读取不到数据为空时, 然后crash了。

```
2020-01-01 00:00:01.423829    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2547), [RADIO], [3], [Tag]RPApnController[0][TAG]: onOperatorNumberChanged (, 4
2020-01-01 00:00:01.424183    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2547), [RADIO], [3], [Tag]RPApnController[0][TAG]: mccmnc.46011.length (5)
2020-01-01 00:00:01.426029    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: generateApnDB.start, this.=0x7
2020-01-01 00:00:01.426042    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: calculate_md5_with_md5sum.comm
2020-01-01 00:00:01.426054    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: calculate_md5_with_md5sum.go.t
2020-01-01 00:00:01.431796    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: calculate_md5_with_md5sum.fget
2020-01-01 00:00:01.431939    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: MD5.of./system/etc/tele/apns-c
2020-01-01 00:00:01.432008    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: last.prop.value.persist.vendor
2020-01-01 00:00:01.451778    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]APN[TAG]: check.init.success!
2020-01-01 00:00:01.452054    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]APN[TAG]: initDb.done!
2020-01-01 00:00:01.452086    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]APN[TAG]: notify.apn.db.ready
2020-01-01 00:00:01.452099    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]APN[TAG]: notify.State.0
2020-01-01 00:00:01.452112    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]APN[TAG]: apn.loading.done!
2020-01-01 00:00:01.452695    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: generateApnDB.APNDB.Done.thi
2020-01-01 00:00:01.452736    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]RPApnController[0][TAG]: GetApnDBBymccmnc.this.=0x7f98
2020-01-01 00:00:01.452749    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [SYSTEM], [3], [Tag]APN[TAG]: apn.update.record.column.(mcc:460)
2020-01-01 00:00:01.452761    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [SYSTEM], [3], [Tag]APN[TAG]: apn.update.irecord.column.(3:11)
2020-01-01 00:00:01.459940    2161 D mtkfusionrild: 0x7faf6c4-0x ( 2161, 2519), [RADIO], [3], [Tag]RmmSimCommUrc[TAG]: [1].handleGsmMccMnc.set.prop.
2020-01-01 00:00:01.465329    2538 D binder:2538_2: 0x7fa9014-0x ( 2538, 2697), [SYSTEM], [3], [Tag]APN[TAG]: do.query.error.no.such.table:apn_table
```

● 问题分析

● 分析日志：SOS\mtklog\mobilelog\APLog_2020_0101_000733_1

从soslog看，第一次开机启动 sos os内核只启动到39秒就重启了。

```
5wmLog_2025_08_30_09_16_17\SOS\mtklog\mobilelog\APLog_2020_0101_000733_1\kernel_log_2020_0101_000733.curf - Notepad++
3(N) 语言(L) 设置(I) 工具(O) 宏(M) 运行(R) 插件(P) 窗口(W) ?
kernel_log_2020_0101_000009.curf x main_log_1_2020_0101_000005 x pl_1k x kernel_log_2020_0101_000733.curf x main_log_2020_0101_000046.curf x
1517 <3>[...38.950662][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Write Fail: cmd=8, addr=0x103, len=0
1518 <3>[...38.986901][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1519 <3>[...39.023217][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1520 <3>[...39.059391][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1521 <3>[...39.095872][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1522 <3>[...39.132400][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1523 <3>[...39.168603][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1524 <3>[...39.204894][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1525 <3>[...39.241277][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1526 <3>[...39.278076][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1527 <3>[...39.315420][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1528 <6>[...39.339530][T1404407].irq/597-b000000:[name:rproc_block&]virtio_block: Programming key @slot1.bitmap=0x1
1529 <6>[...39.341633][T1301069].kworker/u16:5:[name:nebula_log_driver&]nebula: (3) [00047.119] 02555.04514> host.get_vm0.pm_state.1
1530 <3>[...39.351725][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1531 <3>[...39.389893][T1001157].irq/456-11e1000:[name:mediatek_drm&][DPTX][ERROR]Aux Read Fail: cmd=9, addr=0x0, len=1
1532
```

从soslog看，sos内核启动到20秒创建tbox vm1.

```
<6>[ 20.520170][T1600068].kworker/u16:2:[name:nebula_log_driver&]nebula: (5) [00028.297] 02555.04541> notify tee vm created, user_vmid:1 tee_vmid:3
<6>[ 20.537850][T1100068].kworker/u16:2:[name:nebula_log_driver&]nebula: (0) [00028.315] 05408.05427> Create user vmid 1
<6>[ 20.538720][T1100068].kworker/u16:2:[name:nebula_log_driver&]nebula: (5) [00028.316] 05408.05427> [INFO:main_uos.cc(650)] product_id:21 vmid:1 sec_boot:0
<6>[ 20.538728][T1100068].kworker/u16:2:[name:nebula_log_driver&]nebula: (5) [00028.316] 05408.05427> kernel VM3 set user vmid = 1
<6>[ 20.541615][T1101068].kworker/u16:4:[name:nebula_log_driver&]nebula: (0) [00028.319] 02555.02584> [INFO:ipc_service_impl.cc(188)] RegisterIpcEventListener vmid 1 way 1
<6>[ 20.548629][T1503573].nbl_vmm:[name:acrn_nbl&]vm_state_notify: old state=1, new state=3, vmid=1
<6>[ 20.550244][T1600068].kworker/u16:2:[name:nebula_log_driver&]nebula: (3) [00028.324] 02555.02584> [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer_
<6>[ 20.550256][T1600068].kworker/u16:2:[name:nebula_log_driver&]nebula: (3) [00028.324] 02555.02584> [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer_
<6>[ 20.550259][T1600068].kworker/u16:2:[name:nebula_log_driver&]nebula: (3) [00028.324] 02555.02584> [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer_
<6>[ 20.550264][T1600068].kworker/u16:2:[name:nebula_log_driver&]nebula: (3) [00028.324] 02555.02584> [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer_
<6>[ 20.550268][T1600068].kworker/u16:2:[name:nebula_log_driver&]nebula: (3) [00028.324] 02555.02584> [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer_
<6>[ 20.550270][T1600068].kworker/u16:2:[name:nebula_log_driver&]nebula: (3) [00028.324] 02555.02584> [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer_
```

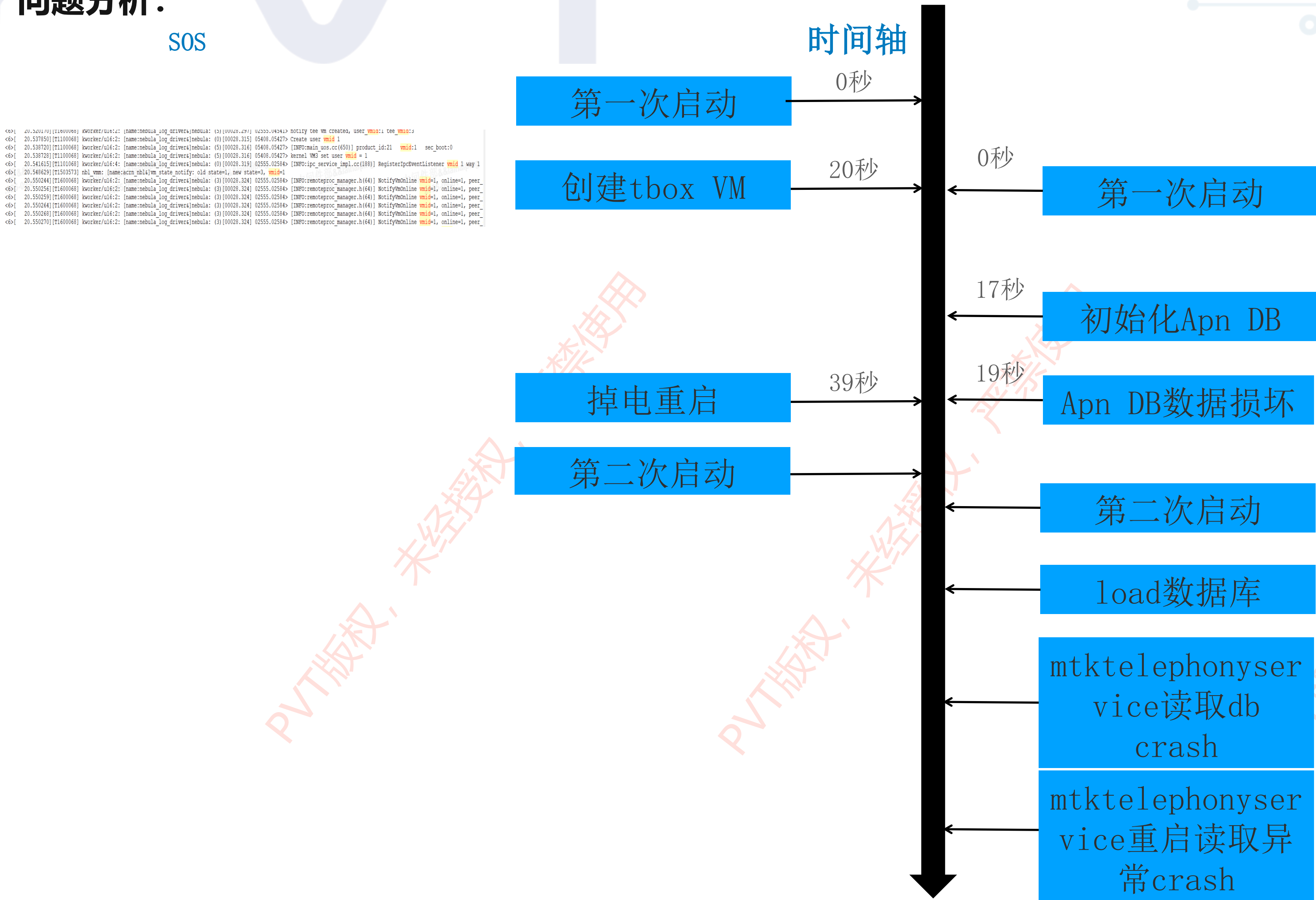
从SOS\mtklog\mobilelog\APLog_2020_0101_000812__2\pl_1k分析第一次重启的原因，发现pmic掉电原因PORSTB，是整机掉电： 外部掉电或者正常 power_off

```
\MT8676\nobo\CUXV5-33265\GwmLog_2025_08_30_09_16_17\SOS\mtklog\mobilelog\APLog_2020_0101_000812__2\pl_1k (匹配1次)
[pmic check rst] PORSTB
```

问题分析：

SOS

Tbox UOS



```
<@> 20.524100 [11000068] kworker/u16:2: [name:nebula_log_driver|nebula: (3) [00028.315] 05408.05427] notify tee vm created, user_vmid:1 tee_vmid:1
<@> 20.537850 [11100068] kworker/u16:2: [name:nebula_log_driver|nebula: (0) [00028.315] 05408.05427] Create user_vmid 1
<@> 20.538720 [11100068] kworker/u16:2: [name:nebula_log_driver|nebula: (5) [00028.316] 05408.05427] [INFO:main.usr.cc(650)] product_id:21 vmid:1 sec_boot:0
<@> 20.538720 [11100068] kworker/u16:2: [name:nebula_log_driver|nebula: (5) [00028.316] 05408.05427] kernel VM3 set user_vmid = 1
<@> 20.541615 [11101068] kworker/u16:4: [name:nebula_log_driver|nebula: (0) [00028.319] 02555.02584] [INFO:ipc_service_impl.cc(188)] RegisterIpcEventListener vmid 1 way 1
<@> 20.548629 [11503573] nbl_wms: [name:acrn_nbl|vm_state_notify: old state=1, new state=3, vmid=1
<@> 20.550244 [11600068] kworker/u16:2: [name:nebula_log_driver|nebula: (3) [00028.324] 02555.02584] [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer
<@> 20.550254 [11600068] kworker/u16:2: [name:nebula_log_driver|nebula: (3) [00028.324] 02555.02584] [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer
<@> 20.550254 [11600068] kworker/u16:2: [name:nebula_log_driver|nebula: (3) [00028.324] 02555.02584] [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer
<@> 20.550264 [11600068] kworker/u16:2: [name:nebula_log_driver|nebula: (3) [00028.324] 02555.02584] [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer
<@> 20.550268 [11600068] kworker/u16:2: [name:nebula_log_driver|nebula: (3) [00028.324] 02555.02584] [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer
<@> 20.550270 [11600068] kworker/u16:2: [name:nebula_log_driver|nebula: (3) [00028.324] 02555.02584] [INFO:remoteproc_manager.h(64)] NotifyVmOnline vmid=1, online=1, peer
```

```
2020-01-01 00:00:01.423829 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2547), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: onOperatorNumberChanged ( 4
2020-01-01 00:00:01.424103 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2547), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: generateApnDB_start, this =>0x7
2020-01-01 00:00:01.424029 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: calculate_mds_with_mdsam com
2020-01-01 00:00:01.424042 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: calculate_mds_with_mdsam get
2020-01-01 00:00:01.424054 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: calculate_mds_with_mdsam get
2020-01-01 00:00:01.431939 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: calculate_mds_with_mdsam get
2020-01-01 00:00:01.432008 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: last prop value persist, vendor
2020-01-01 00:00:01.451778 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]IAPN [TAG]: check init success!
2020-01-01 00:00:01.452086 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]IAPN [TAG]: initdb done!
2020-01-01 00:00:01.452094 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]IAPN [TAG]: notify apn db ready
2020-01-01 00:00:01.452112 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2702), [SYSTEM], [3], [Tag]IAPN [TAG]: apn loading done!
2020-01-01 00:00:01.452695 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: generateApnDB APNDB Done this
2020-01-01 00:00:01.452716 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [RADIO], [3], [Tag]BPAPnController[0] [TAG]: generateApnDB success this =>0x7fa9
2020-01-01 00:00:01.452749 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [SYSTEM], [3], [Tag]IAPN [TAG]: apn update record column (recid:46)
2020-01-01 00:00:01.452761 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [SYSTEM], [3], [Tag]IAPN [TAG]: apn update record column (recid:46)
2020-01-01 00:00:01.455940 2161 D mtktelephony: 0x7fa9014-0x ( 2161, 2519), [RADIO], [3], [Tag]RmsMccmccCtrl [TAG]: (1) handleMccmccCtrl set prop.
2020-01-01 00:00:01.455959 2538 D binder:2538 2: 0x7fa9014-0x ( 2538, 2697), [SYSTEM], [3], [Tag]IAPN [TAG]: do query error no such table apn_label
```

mtktelephonyservice crash异常原因

mtktelephonyservice读取异常的apn db时，取不到apn list，获取指针为空，异常报错

```
void RpApnController::GetApnDBBymccmnc()
{
    logD(RFX_LOG_TAG, " GetApnDBBymccmnc this =%p ,slot_id = %d", this, getSlotId());
    apn_record_t* record = apn_malloc_record();
    if(record != NULL) {
        apn_build_cRecord(record, APN_PROP_MCC, mcc.string());
        apn_build_iRecord(record, APN_PROP_IMNC, mnc.string());
        if(g_apnlist){
            apn_free_list(g_apnlist);
            g_apnlist = NULL;
        }
        apn_query_db(record, &g_apnlist);

        if(g_apnlist == NULL){
            apn_free_record(record);
            //DB_FILE_NAME
            int ret = remove(DB_FILE_NAME);
            if (ret != 0) {
                logE(RFX_LOG_TAG, "delete apn.db failed");
            }
            abort();
            return;
        }
        apn_record_t* instance = g_apnlist->top;
        while(instance != NULL) {
            apn_dump_record(instance);
            instance = (apn_record_t*)instance->next;
        }

        // free all the malloc memory , in case of memory leak
        apn_free_record(record);
        logD(RFX_LOG_TAG, " GetApnDBBymccmnc done,slot_id = %d", getSlotId());
    }
}
```

获取apnlist时，未判断文件损坏获取不到apn list的情况。新增空指针判断，同时删除掉损坏的旧文件，可以确保下次重启能够正常读取。

问题场景模拟：模拟apn db文件损坏场景

1. 删除tbox os内正常的apn db

```
root@auto8676p164uostbox:/# cd /data/vendor/telephony/
root@auto8676p164uostbox:telephony#
root@auto8676p164uostbox:telephony#
root@auto8676p164uostbox:telephony# ls
apn.db
root@auto8676p164uostbox:telephony#
root@auto8676p164uostbox:telephony# rm -rf apn.db
```

2. 本地创建一个乱码的apn db文件push进去

```
PS C:\Users\PVT> adb -s 127.0.0.1:7667 push F:\360MoveData\Users\PVT\Desktop\sos\apn.db /data/vendor/telephony/
F:\360MoveData\Users\PVT\Desktop\sos\apn.db: 1 file pushed, 0 skipped. 0.0 MB/s (16 bytes in 0.003s)
```

3. 重启机器

4. 机器起来查看tbox aee db，模拟发现mtktelephonyservice一直在反复crash

```
root@auto8676p164uostbox:aee_exp# ls
aee.lck  db.00.NE  db.01.NE  db.02.NE  db_history  temp
root@auto8676p164uostbox:aee_exp#
```

5. 查看mtktelephonyservice报错位置，和CUXV5-33265问题报错位置一样。

```
Exception Class: Native (NE)
Exception Type: SIGSEGV

Current Executing Process:
pid: 2555, tid: 2724
/usr/bin/mtktelephonyservice

Backtrace:
#00 pc 000000000013c6b4 /usr/bin/mtktelephonyservice (_ZN15RpApnController16GetApnDBBymccmncEv+132)
#01 pc 000000000013df08 /usr/bin/mtktelephonyservice (_ZN15RpApnController13ApnInit1Thread10ThreadLoopEv+48)
#02 pc 00000000000122dc entry 0000000000000000 /usr/lib64/u/libutils.so.14.0.0 (_ZN7android6Thread11_threadLoopEPv+444)
#03 pc 000000000007ea98 entry 0000000000028590 /usr/lib64/libc.so.6 (pthread_condattr_setshared+1308)
#04 pc 00000000000de908 entry 0000000000028590 /usr/lib64/libc.so.6 (__clone+588)
```

问题场景模拟修复patch验证

1. 增加空指针判断， 以及删除损坏apn db文件。

```
if(g_apnlist == NULL){
    apn_free_record(record);
    //DB_FILE_NAME
    int ret = remove(DB_FILE_NAME);
    if (ret != 0) {
        logE(RFX_LOG_TAG, "delete apn.db failed");
    }
    abort();
    return;
}
```

2. 升级包重启查看mtktelephonyservice是否会一直crash。查看进程pid一直不变，同时看损坏的apn db是否删除。

```
root@auto8676p164uostbox:/# ps -aux | grep mtkte*
root      2548  0.2  1.4 1085360 14320 ?        Sl      2019    0:00 /usr/bin/mtktelephonyservice
root      3184  0.0  0.1   3624  1032 pts/0    S+      00:01    0:00 grep mtkte*
root@auto8676p164uostbox:/# ps -aux | grep mtkte*
root      2548  0.2  1.4 1085360 14320 ?        Sl      2019    0:00 /usr/bin/mtktelephonyservice
root      3186  0.0  0.1   3624  1036 pts/0    S+      00:01    0:00 grep mtkte*
root@auto8676p164uostbox:/# ps -aux | grep mtkte*
root      2548  0.2  1.4 1085360 14320 ?        Sl      2019    0:00 /usr/bin/mtktelephonyservice
root      3188  0.0  0.1   3624  1032 pts/0    S+      00:01    0:00 grep mtkte*
root@auto8676p164uostbox:/# ps -aux | grep mtkte*
root      2548  0.2  1.4 1085360 14320 ?        Sl      2019    0:00 /usr/bin/mtktelephonyservice
root      3190  0.0  0.1   3624  1032 pts/0    S+      00:01    0:00 grep mtkte*
root@auto8676p164uostbox:/#
root@auto8676p164uostbox:/#
root@auto8676p164uostbox:/# ps -aux | grep mtkte*
root      2548  0.2  1.4 1085360 14320 ?        Sl      2019    0:00 /usr/bin/mtktelephonyservice
root      3196  0.0  0.1   3624  1028 pts/0    S+      00:01    0:00 grep mtkte*
```

```
root@auto8676p164uostbox:/# cd data/vendor/telephony/
root@auto8676p164uostbox:telephony#
root@auto8676p164uostbox:telephony# ls
```

3. 重启机器查看网络，网卡正常，ping百度等正常

```
ccmn12: 00-00-00-00-00-00 txqueuelen 1000 (CUNSPEC)
0 frames 0 collisions 0
ccmn13: 00-00-00-00-00-00 txqueuelen 1000 (CUNSPEC)
0 frames 0 collisions 0
ccmn14: 00-00-00-00-00-00 txqueuelen 1000 (CUNSPEC)
0 frames 0 collisions 0
root@auto8676p164uostbox:/# ping www.baidu.com
PING www.a.shifen.com (220.181.111.232) 56(84) bytes of data.
64 bytes from 220.181.111.232: icmp_seq=1 ttl=51 time=39.3 ms
64 bytes from 220.181.111.232: icmp_seq=2 ttl=51 time=46.0 ms
^C
--- www.a.shifen.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 39.254/42.619/45.984/3.365 ms
```

实际场景中，是否会存在这种开机到一半直接下电重启的场景？

结论：实际场景是没有开机到一半掉电的场景。

拉通VIP和Lifecyc的log 证实： 从MCU日志来看，是有进备电模式的。进备电一分钟后，满足重启退出备点条件，因此整机重启退出备电模式。同测试确认是升级完成后，手动掉电重启进入备电模式。

```
1526: 2020-01-01 00:07:24.000 POWER W [16f5cb63210,20200101 00:07:24.104]
行 1527: 2020-01-01 00:07:24.000 POWER W [16f5cb630ed,20200101 00:07:21.581]
行 1627: 2020-01-01 00:07:24.000 POWER D [16f5cb63239,20200101 00:07:21.913]
行 1629: 2020-01-01 00:07:24.000 POWER W [16f5cb63239,20200101 00:07:21.913]
行 1631: 2020-01-01 00:07:24.000 POWER W [16f5cb632f0,20200101 00:07:22.096]
行 1654: 2020-01-01 00:07:24.000 MISC E [16f5cb63742,20200101 00:07:23.202]
行 1660: 2020-01-01 00:07:24.000 MISC E [16f5cb63742,20200101 00:07:23.202]
行 1668: 2020-01-01 00:07:24.000 TC I [16f5cb638a6,20200101 00:07:23.558]
行 1678: 2020-01-01 00:07:24.000 POWER E [16f5cb63960,20200101 00:07:23.744]
行 1678: 2020-01-01 00:07:24.000 POWER E [16f5cb63960,20200101 00:07:23.744]
行 1762: 2020-01-01 00:07:29.000 POWER W [16f5cb64fae,20200101 00:07:29.454]
行 1762: 2020-01-01 00:07:29.000 POWER W [16f5cb64fae,20200101 00:07:29.454]
行 1764: 2020-01-01 00:07:29.000 POWER E [16f5cb64fc2,20200101 00:07:29.474]
行 1766: 2020-01-01 00:07:29.000 POWER E [16f5cb64fc2,20200101 00:07:29.474]
行 1768: 2020-01-01 00:07:29.000 POWER W [16f5cb64fc2,20200101 00:07:29.474]
行 1812: 2020-01-01 00:07:34.000 POWER W [16f5cb66394,20200101 00:07:34.548]
行 1831: 2020-01-01 00:07:39.000 POWER W [16f5cb676bd,20200101 00:07:39.453]
行 1831: 2020-01-01 00:07:39.000 POWER W [16f5cb676bd,20200101 00:07:39.453]
行 1833: 2020-01-01 00:07:39.000 POWER W [16f5cb676de,20200101 00:07:39.478]
行 1851: 2020-01-01 00:08:01.000 POWER E [0,19700101 00:00:00.000]
行 1853: 2020-01-01 00:08:01.000 POWER E [0,19700101 00:00:00.000]
行 1855: 2020-01-01 00:08:01.000 POWER E [0,19700101 00:00:00.000]
行 1857: 2020-01-01 00:08:01.000 POWER E [0,19700101 00:00:00.000]
行 1857: 2020-01-01 00:08:01.000 POWER E [0,19700101 00:00:00.000]
行 1859: 2020-01-01 00:08:01.000 POWER W [0,19700101 00:00:00.000]
行 1877: 2020-01-01 00:08:01.000 POWER E [44,19700101 00:00:00.069]
行 1879: 2020-01-01 00:08:01.000 POWER E [45,19700101 00:00:00.069]
行 1912: 2020-01-01 00:08:01.000 POWER W [47,19700101 00:00:00.071]
行 1915: 2020-01-01 00:08:01.000 POWER W [47,19700101 00:00:00.071]
行 1917: 2020-01-01 00:08:01.000 POWER W [47,19700101 00:00:00.071]
行 1919: 2020-01-01 00:08:01.000 POWER W [47,19700101 00:00:00.071]
行 1921: 2020-01-01 00:08:01.000 POWER W [47,19700101 00:00:00.071]
行 1923: 2020-01-01 00:08:01.000 POWER W [47,19700101 00:00:00.071]
行 1923: 2020-01-01 00:08:01.000 POWER W [47,19700101 00:00:00.071]
730 [2152]get APP types values
731 [2153][442451Ms]livi startup sts:0
767 [2189]FOTA:off,Ensable Sig
768 [2190][442783Ms]MCU got FOTAsts data 0,Send ACK
769 [2191][442966Ms]livi WakeUp all sts:0
780 [2202]log level set POWER 5
783 [2205]log level set SOCM 5
787 [2209]HMI power status = 0
792 [2214]Power Voltage Hold At 12.18v
792 [2214]Power Voltage Hold At 12.18v
832 [2254]hidden power trigger lmin alive countdown:20
832 [2254]hidden power trigger lmin alive countdown:20
833 [2255]BackBatter error High:ID_EN
834 [2256]BackBatter error High:CD_EN
835 [2257]hidden energy working mode!
856 [2278][455418NM state:2, MCUAlive state:1, IGN state:0]
864 [2286]hidden power trigger lmin alive countdown:10
864 [2286]hidden power trigger lmin alive countdown:10
865 [2287]hidden energy working mode!
3 [4]Wakeup flag WuF0 is : 0
4 [5]Wakeup flag WuF1 is : 0
5 [6]RESF is HEX: 1
6 [7]hidden power mode recovery reset!!!!
6 [7]hidden power mode recovery reset!!!!
7 [8]System Warm Reset! RestType:0
16 [17]ADC SelfDiag OK!!!,AWOVCL 4090 ISOVCL 4090
17 [18]ADC OpenDiag OK!!!,AWOVCL 1024 ISOVCL 1023
31 [32][71Ms]SM Voltage Origin->VOLT Normal
32 [33][71Ms]SM Origin->SP_STATE SP Off;SP trans ID:0
33 [34]SM Origin->IVI_OFF;IVI transID:0
34 [35]RemoteFlag = 0;OTA UpgradeModStFlag = 0
35 [36][71Ms]SM Origin->Cluster_OFF;Cluster transID:0
36 [37]Power_Init finished
36 [37]Power_Init finished
```

备电状态计时一分钟后，满足重启条件
车机重启退出备电模式

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